

Measuring Muscle, Automatically

Geometry-aware deep learning for muscle architecture
estimation in B-mode ultrasound

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CSE 754 Deep Learning for Computer Vision

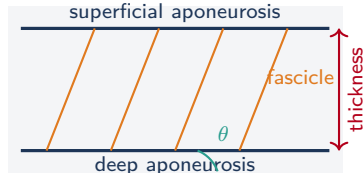
The problem

Three numbers describe muscle architecture:

- **muscle thickness** (mm)
- **pennation angle** (degrees)
- **fascicle length** (mm)

They are still measured **by hand**: slow, subjective, and dependent on operator experience.

Goal. Produce all three automatically from a single ultrasound frame.



What the task actually is

The data does not make this a supervised regression problem.

No measurement labels

Training data gives *masks* only, never a thickness or an angle. The three targets must be **reconstructed geometrically**.

No scale metadata

Two targets are in millimetres; the network sees pixels. The conversion must be **read off the scanner's own depth ruler**.

Silent data defects

- 70% of frames disagree in size with their own mask
- 26% of fascicle images are duplicates

Neither raises an error. Both corrupt results in ways that look like success.

A naive split leaks **179** duplicate groups into validation. Content-grouped splitting reduces that to **0**.

Pipeline



Data

1 048 aponeurosis frames
2 761 fascicle frames
309 unlabelled test frames

Model

U-Net, ResNet-34 encoder
512×512, BCE + Dice
94 min on one GPU

Contribution

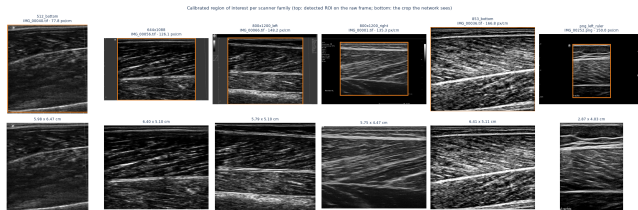
scale recovery
dense orientation head
geometry in mm space

Recovering physical scale

Six scanner families, each drawing its depth ruler differently: left edge, right edge, bottom edge; ticks every 5 or 10 mm.

100% of the 309 test frames calibrate

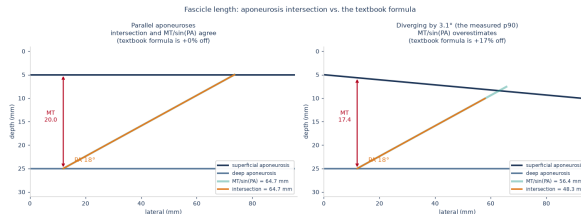
Validated independently: implied field of view spans **2.97** to **6.95** cm deep, matching the documented 3 to 7 cm scanner presets. Nothing was fitted to those numbers.



Detected region of interest per scanner family, and the crop the network sees.

Geometry: from masks to millimetres

- Aponeuroses fitted as **curves**, not lines
- Pennation referenced to the **deep aponeurosis**, as the clinical definition specifies
- Orientation predicted as a dense **$(\cos 2\theta, \sin 2\theta)$ field**, so that averaging directions is well defined



Two ways to reconstruct fascicle length. Which is better turned out to be the project's most interesting question.

Orientation targets are *derived*, so they were checked against an independent estimator: median disagreement **1.1** degrees, 100% of frames within 5 degrees.

Evaluating without any measurement labels

No frame carries an expert measurement, so error against a human cannot be computed anywhere.

Pseudo-ground-truth protocol. Run the *same* geometry module twice, once on the annotator's masks and once on the network's, and compare.

What it measures

the error the segmentation contributes, with the geometry held fixed.

What it does not measure

error of the geometry module itself against a human. That needs expert annotations this data does not contain.

External check. Pennation angle is scale-invariant, so the reconstruction run on annotator masks can be compared with the outside world without trusting the calibration: it gives a median of **17.0** degrees, against a published range of 10 to 25 degrees and **16.4** degrees for an independently built pipeline.

Segmentation, held out

	aponeurosis	fascicle
Dice	0.839	0.274
IoU	0.731	0.161
Orientation		2.74°

The fascicle Dice is low because annotators traced a *representative sample* of fascicles, not all of them. A model finding a different valid set is penalised while being correct for the purpose.

Geometry, predicted vs annotator masks

Muscle thickness	0.3% relative
Pennation angle	1.16°
Fascicle length	6.0% relative

Thickness is recovered from *predicted* masks to within 0.3% of what the annotator's own masks give. That single fact drove the project's main design decision.

Caveat: all 78 paired frames come from one scanner family.

Three of four proposed mechanisms did not earn their place

Proposed	Measured against what it replaced
Dense orientation head	Does not beat it. 0.79° against 0.72° for the classical structure tensor.
Sequence smoothing	Nearly inert. Within-run spread was already 0.02 mm in thickness.
Per-frame confidence	Does not predict error. Spearman -0.08 to -0.19 , not significant.
Intersection for fascicle length	Helps, but not for the proposed reason. See next slide.

The confidence measure initially *appeared* to detect failures. It was being told about them: an earlier version docked confidence whenever an estimate hit a clamp, so any correlation with error was circular. The term was removed.

The main finding

A model that is more faithful in principle can be a worse estimator in practice, when it depends on noisier inputs.

The proposal argued that $FL = MT/\sin(PA)$ is wrong because the aponeuroses are not parallel, and that intersecting the fascicle with the fitted curves is better. The premise is correct. The conclusion does not follow.

Construction	Median error
Intersection	0.077
Textbook $MT/\sin(PA)$	0.060

Wilcoxon $p = 0.0003$, $n = 78$.

Why. Not because divergence is estimated badly: it is recovered to 0.15° . The textbook formula leans on *thickness*, recovered to 0.3%. The intersection leans on a *difference of two fitted angles*, inheriting angular noise twice over.

What the intersection actually contributes is **boundedness**: it cannot diverge at shallow pennation, where $MT/\sin(PA)$ runs away as $1/\sin$. Right mechanism, wrong reason.

What worked

- Thickness to **0.3%**, pennation to **1.16°**
- Aponeurosis Dice **0.839**
- Scale recovered on **100%** of test frames

The component that mattered most was **scale recovery**, which the original proposal never contemplated, because it assumed the measurements were learnable targets.

Honest limitations

- No expert-measurement validation anywhere
- Paired set is 78 frames from one scanner family
- Fascicle length is extrapolated, not observed
- Single split, single seed

Code, report and full ablations: github.com/silvererudite/umud-muscle-architecture